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GEARGENIX: Advanced Campus Gear Control

A Synopsis for Minor Project

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

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◦ Introduction

Today's colleges face a critical resource management crisis, as over 20 clubs and departments compete for limited shared equipment like projectors, microphones, laptops, and speakers. This leads to chaotic WhatsApp coordination, frequent double bookings, zero accountability, and complete visibility gaps during peak event seasons. GearGenix delivers a revolutionary AI-powered solution to this problem through a conversational agent utilizing OpenAI with the ReAct pattern and function calling.

Users can simply send a message such as "Book projector tomorrow 2-4 PM for Robotics Club" via a Telegram bot or a React web interface. The intelligent agent instantly checks real-time availability, detects conflicts using proven interval overlap logic, and manages the complete booking lifecycle. Built on a production-grade stack comprising a FastAPI backend and PostgreSQL database, this scalable system eliminates disorder, ensures transparency, and empowers over 1000 users with seamless, natural language resource management.

◦ Literature Review

Existing research reveals significant gaps in current college equipment management systems. Traditional web-based reservation platforms such as e-Bosy and Cheqroom focus heavily on form-based interfaces [1], [4]. However, they lack conversational AI capabilities, forcing users through rigid workflows that colleges find cumbersome. Academic studies document persistent problems; for example, the University of Cape Town reported systemic double-booking failures in centralized venue systems, while James Cook University highlighted manual equipment tracking inefficiencies [2]. Both instances lacked real-time AI conflict resolution.

A critical gap identified in current literature is that no existing system combines conversational AI with intelligent conflict resolution for campus resource sharing. Current solutions remain rigid, manual, or domain-specific without natural language understanding and real-time orchestration. GearGenix serves as an innovation in this space by being among the first system to deliver AI-driven conversational booking with tool-chaining (check, book, confirm) specifically for higher education equipment sharing, directly addressing the 85% coordination failure rate documented across 13 institutions [2]. GearGenix builds upon proven ReAct patterns [3] while solving a unique coordination problem absent from current literature.

◦ Problem Definition

In colleges with 20+ clubs and departments, shared equipment creates systemic coordination failures. The core problems identified include:

- **No Centralized System:** WhatsApp groups replace structured booking processes.
- **Double Bookings:** A lack of real-time availability tracking results in an 85% failure rate.
- **Zero Accountability:** There are no concrete records of who is in possession of what equipment.
- **Manual Faculty Dependency:** One-to-one inquiries overload the teaching staff.
- **Event Season Collapse:** Peak demand exposes the complete failure of the current manual system.

The quantified impact of this disorganized current state—where faculty inquiries move to

WhatsApp and rely on memory-based tracking—results in 3 to 5 hours wasted daily per club on manual coordination. Consequently, 70% of events are delayed or disrupted due to double bookings, and faculty spend over 15 hours a week managing equipment queries.

◦ Objectives

Primary Goals:

- **Eliminate Double Bookings:** Deploy real-time conflict detection to achieve 100% booking accuracy.
- **Replace Manual Chaos:** Enable natural language interaction through a Telegram bot and React web application.
- **Centralize Management:** Provide a single platform for all 20+ clubs with complete visibility.

Technical Objectives:

- Build an AI conversational agent using the OpenAI ReAct pattern with function calling.
- Implement 8 core functionalities: list, check, book, view, cancel, return, and admin view.

Operational Objectives:

- **Zero faculty dependency:** Achieve autonomous management of the booking lifecycle.
- **Complete transparency:** Ensure real-time equipment status is visible to all stakeholders.
- **Multi-platform access:** Provide a seamless experience across Telegram and web platforms.

◦ Methodology

The project is structured into four distinct development phases:

- **Phase 1: Foundation (Database Design)**
 - Implement a PostgreSQL relational schema utilizing equipment and bookings tables.
 - Use foreign key relationships to ensure data integrity.
 - Rely on ACID transactions to guarantee conflict-free concurrent bookings.
- **Phase 2: AI Agent Architecture**
 - Utilize the OpenAI ReAct Pattern (Observe → Reason → Act → Repeat).
 - Integrate Function Calling with 7 intelligent tools, such as `list_equipment` and `make_booking`.
 - Maintain conversation memory by keeping the full chat history for multi-turn context.
- **Phase 3: Interface Layer**
 - Develop a Telegram Bot using `python-telegram-bot` for instant mobile accessibility.

- Create a Web Dashboard using a React and Vite frontend paired with FastAPI REST endpoints.
- Route all conversations to the AI agent via a unified /chat endpoint.
- **Phase 4: Core Intelligence**
 - Implement a conflict detection algorithm using the logic $\text{start1} < \text{end2} \text{ AND } \text{end1} > \text{start2}$ to catch all overlaps.
 - Use Tool Chaining so the agent can auto-execute `check_availability` followed by `make_booking`.
 - Generate human-readable booking IDs, such as B001 and B002.

○ **References**

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